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A THREE-NODE MINDLIN PLATE ELEMENT WITH IMPROVED TRANSVERSE SHEAR

Alexander TESSLER
Mechanics of Materials Branch, Army Materials and Mechanics Research Center, Watertown, MA 02172, U.S.A.

and

Thomas J.R. HUGHES*
Division of Applied Mechanics, Stanford University, Stanford, CA 94305, U.S.A.

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A displacement methodology for Mindlin elements, recently employed in the development of an efficient, four-node quadrilateral (MIN4), is the basis for a three-node, explicitly integrated triangular element (MIN3). The approach, herein referred to as *anisoparametric*, is founded upon two kinematic requirements that are auxiliary to the standard convergence criteria. The initial, complete quadratic deflection field is constrained by 'continuous' shear edge constraints to achieve an isoparametric-like, three-node configuration. Much success of the element is due to an innovative shear correction factor concept. Several numerical experiments are performed in the linear elasto-static regime to ascertain the element performance.

1. Introduction

Computational advantages of simple plate and shell elements in large-scale finite element analysis and, especially, in nonlinear problems, have motivated extensive development of three-node triangles and four-node quadrilaterals (e.g., see [1–18]). Much effort concentrated on utilizing a refined (i.e., shear-deformable) bending theory of Mindlin [19], where only C^0 kinematic continuity is needed, and the range of applicability includes moderately thick plates. An additional attraction of this approach is that an analogous theory for laminated composites due to Yang, Norris and Stavsky [20] permits a convenient methodology transfer to composite plate elements.

Regrettably, simple Mindlin elements are commonly generated from lower-order interpolations (i.e., bilinear or linear) that tend to be insufficient for the bending action. Using a standard isoparametric approach (i.e., the same shape functions for all kinematic variables and the geometry, and full integration of all energy terms), there result, in the thin bending regime, solutions that may be orders of magnitude stiffer than the true ones (e.g., see discussions in [8, 21, 22]). This mathematical pitfall is often referred to as 'locking'. To alleviate this

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deficiency, a number of variationally inconsistent but practically enhancing special devices have been tried (see, for example, [11, 15, 21] for discussions of these techniques and references to pertinent literature). Despite the general behavior improvements achieved by the ad hoc schemes, there still remain drawbacks. The most common ones are poor stress recovery, drastic deterioration in accuracy under geometric distortions, inferior accuracy under distributed static and dynamic (inertial) loadings due to inconsistent load vectors and inconsistent mass matrices. Furthermore, some elements suffer from rank deficiency [8, 16, 22].

In [15], Tessler and Hughes developed a conforming, four-node (twelve degrees-of-freedom) quadrilateral element (MIN4) based upon a displacement formulation of Mindlin plate theory. Recognizing the transverse shear energy as the major cause of difficulty under isoparametric kinematic assumptions, a special interpolation scheme was devised. Additionally, the element transverse shear energy was further enhanced by a suitable (element-appropriate) shear correction factor. These measures, in conjunction with full numerical integration, produced a well-conditioned, correct rank stiffness matrix accompanied by variationally consistent load vector and mass matrix. As a result of the extensive numerical testing [15], MIN4 evolved as one of the best performing four-node elements and is virtually devoid of the above-mentioned deficiencies. It was suggested in [15] that, using essentially the same methodology, an efficient three-node (nine degrees-of-freedom) triangular element (MIN4 compatible) could be developed. It is the purpose of this effort to derive such an element (herein referred to as MIN3) and test it extensively in the linear, elasto-static plate regime.

In Section 2, we discuss the shear locking phenomenon in relation to a linear three-node isoparametric triangle, and point toward an interpolational inadequacy of this approach. We then propose two auxiliary kinematic criteria impelled by transverse shear (penalty) constraints (Section 3), on which basis an interpolation strategy, herein referred to as *aniso-parametric*, is introduced (Section 4). The explicit edge constraints, of the kind employed in [15], are used to arrive at a three-node, isoparametric-like nodal configuration (Section 4). The element matrices, obtained by explicit exact quadratures, and the stress resultants, are summarized in Section 5. An expanded presentation of the element-appropriate shear correction factor concept is attempted in Section 6 in a somewhat more general manner than in [15]. A phenomenon of overrestraining of a kinematic field by boundary restraints is elaborated upon in Section 7 using a simple two-node beam example.

Numerical experiments with MIN3 for thin and moderately thick plates in the elasto-static regime are presented in Section 8. Moreover, MIN4 is further tested herein to assess its performance under skew distortions. Finally, concluding remarks are presented in Section 9.

2. Isoparametric shear locking

To ascertain the cause of difficulties associated with a Mindlin plate formulation, let us first examine a strain energy functional for a linearly elastic, isotropic plate element of thickness h, area A, Young's modulus E and Poisson's ratio ν (refer to Fig. 1 for the plate notation), i.e.,

$$U = \tfrac{1}{2}D\Big\{\int\int [\theta_{y,x}^2 + 2\nu\theta_{y,x}\theta_{x,y} + \theta_{x,y}^2 + \tfrac{1}{2}(1-\nu)(\theta_{x,x} + \theta_{y,y})^2]\,\mathrm{d}A + \frac{\alpha}{A}\int\int (\gamma_{xz}^2 + \gamma_{yz}^2)\,\mathrm{d}A\Big\}, \tag{2.1}$$

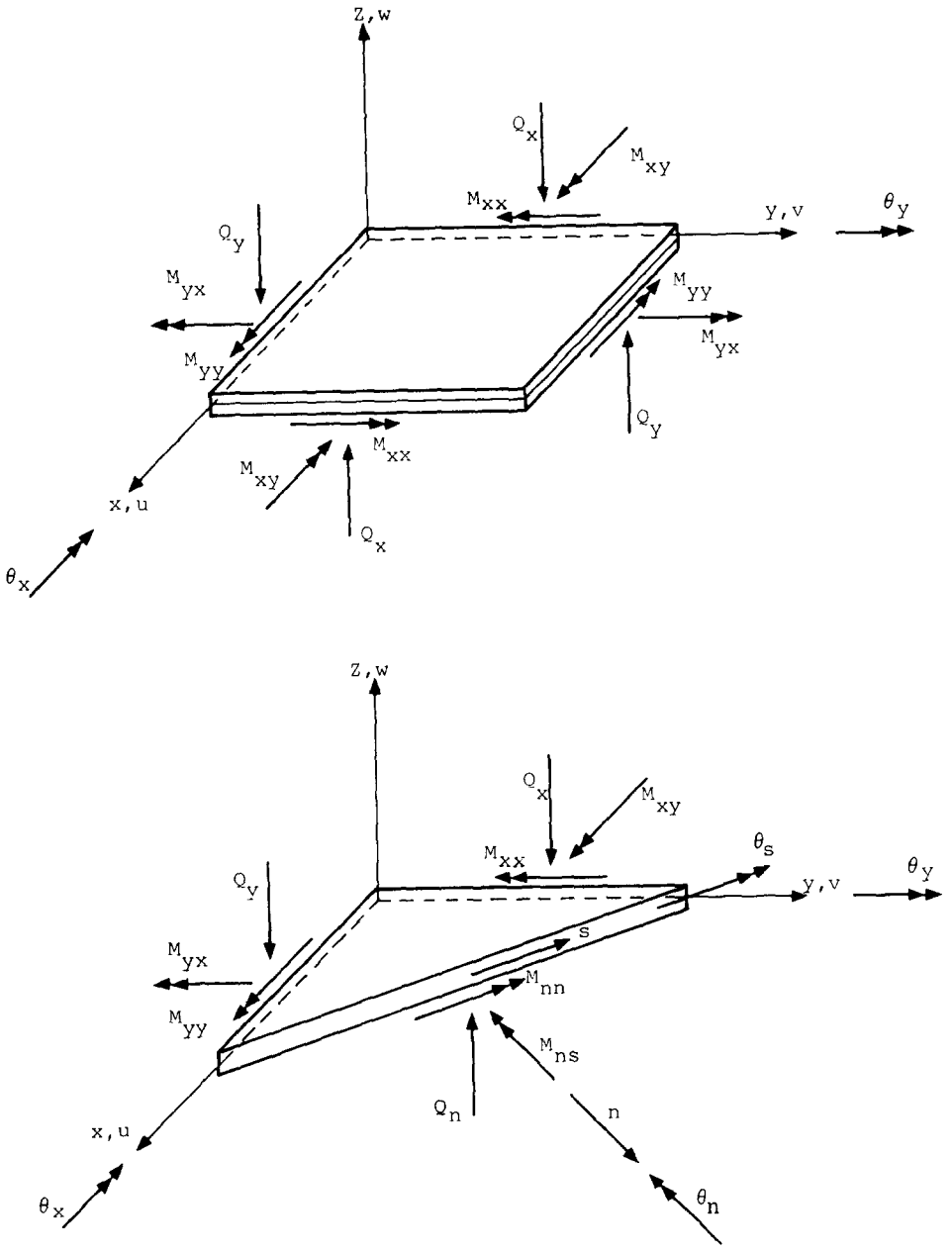


Fig. 1. Plate notations: kinematic variables and stress resultants.

where the first and second integrals represent nondimensional bending and transverse shear strain energies, respectively; $D = Eh^3/12(1-\nu^2)$ is the bending rigidity; $\alpha = 6k^2(1-\nu)A/h^2$ is the penalty parameter; k^2 is the shear correction factor, and

$$\gamma_{xz} = w_{,x} + \theta_y, \qquad \gamma_{yz} = w_{,y} + \theta_x \tag{2.2}$$

are the transverse shear (penalty) strains; a comma is used to denote partial differentiation.

The second integral in (2.1) may be interpreted as a penalty constraint functional. Its purpose is twofold: (1) to enforce the classical limit of vanishing transverse shear as the plate approaches its thin limit (i.e., as $h \to 0$ the Kirchhoff constraints γ_{xz}, $\gamma_{yz} \to 0$ are enforced); (2) to accommodate the non-classical (thick) regime accounting for the transverse shear effect.

It is emphasized that the limiting classical regime is critical for element behavior. As the penalty parameter tends to infinity as $h \to 0$, it is desirable to ensure that the vanishing shear strains do not originate any spurious constraining. This is because spurious constraining, often referred to as 'locking', constitutes a major stumbling block leading to excessively stiff and essentially useless solutions.

Nearly all isoparametric elements (i.e., elements employing the same parametric functions for the deflection, w, and normal rotation, θ_x and θ_y, variables, see e.g. [21]), and especially the lower-order ones, suffer from the locking effect under full integration of the stiffness matrix (the term 'full integration' implies a minimum-order numerical (usually Gauss) quadrature rule that produces exact integrals taken over a rectangular or triangular plan). At first glance, it appears paradoxical that such an approach leads to worthless results even though all of the fundamental convergence requirements are fulfilled (i.e. rigid body modes, constant strain, compatibility and differentiability criteria [21]). However, upon examination of the penalty strains while under Kirchhoff constraints, the cause of the spurious constraining becomes apparent.

To illustrate, consider an isoparametric, linear triangular element located in the x-y Cartesian framework whose origin is at the element centroid. The kinematic interpolations may be written as

$$w = c_{00} + c_{10}x + c_{01}y\,, \tag{2.3a}$$

$$\theta_i = d_{00}^{(i)} + d_{10}^{(i)}x + d_{01}^{(i)}y \quad (i = x, y)\,, \tag{2.3b}$$

where c_{kl} and $d_{kl}^{(i)}$ depend upon the w_j and θ_{ij} $(1 \leqslant j \leqslant 3)$ degrees-of-freedom (dof), respectively. The Kirchhoff constraints

$$\begin{aligned} 0 \leftarrow \gamma_{xz} &= w_{,x} + \theta_y \\ &= (c_{10} + d_{00}^{(y)}) + d_{10}^{(y)}x + d_{01}^{(y)}y \end{aligned} \tag{2.4}$$

and

$$\begin{aligned} 0 \leftarrow \gamma_{yz} &= w_{,y} + \theta_x \\ &= (c_{01} + d_{00}^{(x)}) + d_{10}^{(x)}x + d_{01}^{(x)}y \end{aligned} \tag{2.5}$$

are enforced in the limit as $h \to 0$. If the shear strain energy is integrated exactly, which is in accordance with variational requirements, then all six constraints resulting from (2.4), (2.5) are approximately enforced. Among these, the four constraints

$$d_{10}^{(y)}, d_{01}^{(y)}, d_{10}^{(x)}, d_{01}^{(x)} \to 0\,, \tag{2.6}$$

exclusively dependent on the rotational dof, are the spurious (nonphysical) ones. Upon their enforcement, normal rotations (2.3b) can assume either zero or constant values, and this

causes the bending energy to vanish (see (2.1)). The practical implication is that excessively stiff results are obtained, i.e., the solution is 'locked' due to spurious shear constraining.

To alleviate the spurious locking phenomenon within the framework of the isoparametric assumptions, a one-point centroidal quadrature on the shear strain energy might be used, which effectively 'drops' the spurious constraints (i.e., (2.4), (2.5) are evaluated at $x = y = 0$.) Such reduced integration schemes, though often plagued by one or more spurious zero energy modes, have been used widely due to their simplicity and beneficial effects (e.g., refer to [8, 12, 17, 22]).

The locking phenomenon can also be linked to the inability of the isoparametric, exactly integrated elements to represent the state of constant shear, even though a constant term is present in each shear strain. To observe this, consider a simple one-dimensional example – a cantilevered beam loaded by a shear force at the tip. Assuming the beam axis coincides with the x-direction and setting $y = 0$ (i.e., considering a beam element), it becomes apparent that the theoretically constant shear state ($\gamma_{xz} = \text{const.}$) can only be achieved if $d_{10}^{(y)} = 0$. However, in the linear isoparametric element,

$$d_{10}^{(y)} \propto \theta_{y1} - \theta_{y2} \tag{2.7}$$

and for $d_{10}^{(y)}$ to vanish, the two rotational dof must be equal or be identically zero. But in this case, again, the bending energy vanishes and locking is encountered. On the other hand, a non-locking solution can be obtained in the thick regime [8], in which case $\theta_{y1} \neq \theta_{y2}$, yielding a linear (not constant!) shear.

It then follows that the isoparametric approach should be regarded inadequate for the treatment of the penalty functional of (2.1).

3. Auxiliary criteria

In order to produce effective non-locking solutions, it is proposed that the kinematic field adheres to the auxiliary, penalty-constraint impelled requirements:

CRITERION 3.1. Constant penalty strains must be accommodated when such deformation states are imposed upon a finite size element.

CRITERION 3.2. The classical limiting condition of vanishing shear strains must be attainable without engendering any spurious constraining.

These criteria point toward the notion of 'proper' shear strain polynomials [23]. In these polynomials, each generalized coordinate is represented by a linear combination of the deflection and normal rotation dof, i.e., they possess no spurious terms dependent on a single independent variable. It can be readily ascertained that the proper, complete polynomials are perfectly suited for triangular elements (also one-dimensional elements such as beams and axisymmetric shells, see [24, 25]). Such shear strain polynomials of degree p may be expressed as

$$\gamma_{jz}(w_{,j};\theta_i)\overset{\text{df}}{=}\sum_{\alpha=0}^{p}\sum_{\beta=0}^{p}a_{\alpha\beta}^{(j)}x^{\alpha}y^{\beta}, \tag{3.1a}$$

$$a_{\alpha\beta}^{(j)}=0 \quad \forall(\alpha+\beta)>p, \qquad \alpha_{\alpha\beta}^{(j)}\neq 0 \quad \forall(\alpha+\beta)\leq p, \tag{3.1b}$$

$$a_{\alpha\beta}^{(j)}=a_{\alpha\beta}^{(j)}(w_k,\theta_{ik}) \quad (j=x,y;\ i=y,x;\ i\neq j,\ p=1,2,\ldots), \tag{3.1c}$$

where w_k and θ_{ik} denote the appropriate nodal dof.

The form of (3.1) suggests that the polynomial for w must be one degree higher than that for θ_i. Hence, the desired triangular element interpolations can be written as

$$w=\sum_{\alpha=0}^{p+1}\sum_{\beta=0}^{p+1}c_{\alpha\beta}x^{\alpha}y^{\beta}, \qquad \theta_i=\sum_{\alpha=0}^{p}\sum_{\beta=0}^{p}d_{\alpha\beta}^{(i)}x^{\alpha}y^{\beta},$$

and

$$w_{,j}=\sum_{\alpha=0}^{p}\sum_{\beta=0}^{p}c_{\alpha\beta}^{(j)}x^{\alpha}y^{\beta} \quad (j=x,y;\ \iota=y,x;\ i\neq j)$$

$$\gamma_{jz}=w_{,j}+\theta_i=\sum_{\alpha=0}^{p}\sum_{\beta=0}^{p}a_{\alpha\beta}^{(j)}x^{\alpha}y^{\beta}, \quad a_{\alpha\beta}^{(j)}=c_{\alpha\beta}^{(j)}+d_{\alpha\beta}^{(i)}, \tag{3.2}$$

where conditions on $c_{\alpha\beta}$, $c_{\alpha\beta}^{(j)}$ and $d_{\alpha\beta}^{(i)}$ are the same as those on $a^{(j)}$ in (3.1). Consequently, the deflection slopes $w_{,j}$ are represented by the same complete polynomials as the normal rotations θ_i.

It must be remarked that fulfillment of Criterion 3.1 will generally guarantee satisfaction of Criterion 3.2 as well, unless excessive boundary restraints are prescribed. One such circumstance will be demonstrated on a Timoshenko beam example (Section 7).

4. Anisoparametric interpolations

Interpolations (3.2), once expressed in terms of the element parametric coordinates, will be referred to as *anisoparametric*. The term is meant to emphasize the distinct order of the kinematic interpolations. The well-established isoparametric approach would then constitute a sub-class of the anisoparametric strategy.

In the derivation of a triangular element, it is convenient to utilize parametric coordinates expressed in terms of subtriangle areas (refer to Fig. 2), i.e.,

$$(\xi_1,\xi_2,\xi_3)=\frac{1}{A}\{A_1,A_2,A_3\}, \quad \sum_{i=1}^{3}A_i\equiv A. \tag{4.1}$$

The unique linear relation between Cartesian and parametric coordinates,

$$(1,x,y)=\left\{\sum_{i=1}^{3}\xi_i,\sum_{i=1}^{3}\xi_i x_i,\sum_{i=1}^{3}\xi_i y_i\right\}, \tag{4.2}$$

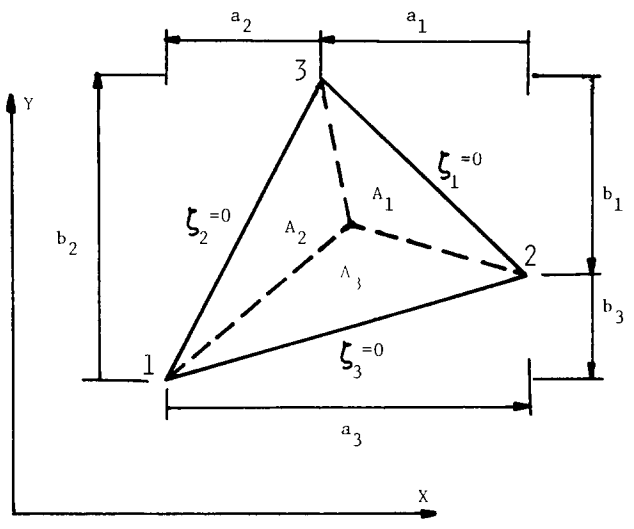


Fig. 2. Triangular element coordinate description.

permits an inverse relation as well. The simple integration formula

$$\int\int \xi_1^a \xi_2^b \xi_3^c \, dx\, dy = \frac{a!b!c!}{(a+b+c+2)!} 2A \tag{4.3}$$

allows for explicit exact quadratures of all element matrices.

The initial kinematic assumptions for the lowest interpolation order element ($p = 1$) are taken as (refer to Fig. 3 for the nodal pattern):

quadratic deflection

$$w = \boldsymbol{N}\boldsymbol{w}_v, \tag{4.4}$$

where the row vector of quadratic trial functions and the associated vector of nodal dof are

$$\boldsymbol{N} = \{N_i\}, \qquad \boldsymbol{w}_v^t = \{w_i\} \quad (i = 1, \ldots, 6),$$

with

$$N_i = \xi_i(2\xi_i - 1), \qquad N_{i+3} = 4\xi_i\xi_k \quad (i = 1, 2, 3;\ k = 2, 3, 1); \tag{4.4a}$$

linear rotations

$$\theta_i = \boldsymbol{\xi}\boldsymbol{\theta}_i \quad (i = x, y), \tag{4.5}$$

where the row vector of linear trial functions and the associated vectors of nodal dof are

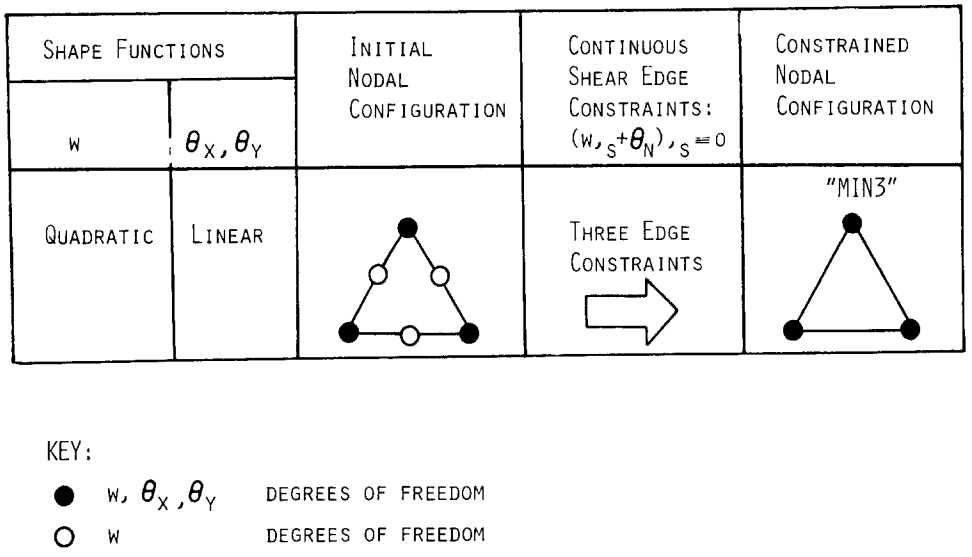


Fig. 3. Anisoparametric triangular element; nodal configurations for initial (unconstrained) and constrained displacement field.

$$\boldsymbol{\xi} = \{\xi_k\}, \qquad \boldsymbol{\theta}_i^{\mathrm{t}} = \{\theta_{ik}\} \quad (i = x, y;\ k = 1, 2, 3)\,. \tag{4.5a}$$

Equations (4.4), (4.5) can be directly used in formulating element matrices. However, it may be advantageous, from the standpoint of nodal simplicity, to condense out the mid-edge deflection dof. This can be accomplished following procedures developed in [15, 24], namely: enforce *continuous* shear constraints at every element edge as given by the differential relation

$$\gamma_{sz,\,s} = (w_{,\,s} + \theta_n)_{,s}|_{\xi_i=0} \equiv 0 \quad (i = 1, 2, 3)\,, \tag{4.6}$$

where s denotes the edge coordinate and θ_n is the tangential edge rotation (refer to Fig. 1). The enforcement of constraints (4.6) at the three element edges yields

$$w_{i+3} = \tfrac{1}{2}(w_i + w_j) + \tfrac{1}{8}[b_k(\theta_{xi} - \theta_{xj}) + a_k(\theta_{yj} - \theta_{yi})] \quad (i = 1, 2, 3;\ j = 2, 3, 1;\ k = 3, 1, 2)\,. \tag{4.7}$$

Upon substituting (4.7) into w in (4.4), there results a *constrained* deflection field exclusively in terms of vertex dof (refer to the nodal patterns shown in Fig. 3), i.e.,

$$\begin{aligned} &w = \boldsymbol{\xi}\boldsymbol{w} + \boldsymbol{L}\boldsymbol{\theta}_x + \boldsymbol{M}\boldsymbol{\theta}_y\,,\\ &\boldsymbol{L} = \{L_i\}, \qquad \boldsymbol{M} = \{M_i\}, \qquad \boldsymbol{w} = \{w_i\}, \qquad \boldsymbol{\theta}_x = \{\theta_{xi}\}, \qquad \boldsymbol{\theta}_y = \{\theta_{yi}\}\,,\\ &L_i = \tfrac{1}{8}(b_k N_{i+3} - b_j N_{k+3}), \qquad M_i = \tfrac{1}{8}(a_j N_{k+3} - a_k N_{i+3})\\ &\quad (i = 1, 2, 3;\ j = 2, 3, 1;\ k = 3, 1, 2)\,, \end{aligned} \tag{4.8}$$

where the conditions

$$\sum_{i=1}^{3} L_i \equiv 0, \qquad \sum_{i=1}^{3} M_i \equiv 0, \qquad \sum_{i=1}^{3} \xi_i \equiv 1 \tag{4.9}$$

guarantee satisfaction of the fundamental constant strain criterion.

The deflection given by (4.8) possesses continuity of order 0 (C^0) across element edges. Together with the linear rotation assumptions (4.5), this constrained deflection field of the second degree will be employed in the formulation of the element matrices. The element will be referred to as MIN3 (Mindlin-type, three nodes).

5. Element matrices and stress resultants

In this section we proceed with the description of the element stiffness matrix and consistent load vector for a linearly elastic, anisotropic plate element of constant thickness h. The derivation of these matrices is straightforward using either the virtual-work or the principle of minimum potential eergy approaches (e.g., refer to [15, 21]). Extensions to fully nonlinear analysis may be achieved by procedures described in [26].

The element stiffness matrix, $\boldsymbol{K}$, may be expressed in terms of its bending, $\boldsymbol{K}_b$, and transverse shear, $\boldsymbol{K}_s$, components as

$$\boldsymbol{K} = \boldsymbol{K}_b + \boldsymbol{K}_s$$

$$= \int_A \int \boldsymbol{B}_b^t \boldsymbol{D}_b \boldsymbol{B}_b \, dA + \int_A \int \boldsymbol{B}_s^t \boldsymbol{G}_s \boldsymbol{B}_s \, dA, \tag{5.1}$$

where

$$\boldsymbol{B}_b = \begin{bmatrix} 0 & 0 & \xi_{,x} \\ 0 & \xi_{,y} & 0 \\ 0 & \xi_{,x} & \xi_{,y} \end{bmatrix}, \tag{5.1a}$$

$$\boldsymbol{B}_s = \begin{bmatrix} \xi_{,x} & L_{,x} & M_{,x} + \xi \\ \xi_{,y} & L_{,y} + \xi & M_{,y} \end{bmatrix}. \tag{5.1b}$$

The bending and transverse shear rigidity matrices are, respectively,

$$\boldsymbol{D}_b = \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ & D_{22} & D_{26} \\ \text{sym.} & & D_{66} \end{bmatrix}, \qquad \boldsymbol{G}_s = \begin{bmatrix} G_{11} & G_{12} \\ \text{sym.} & G_{22} \end{bmatrix}, \tag{5.1c}$$

where

$$D_{ij} = \tfrac{1}{12} h^3 C_{ij}, \qquad G_{ij} = k_{ij}^2 h C_{6-i,\,6-j},$$

with C_{ij} denoting the plane stress elastic moduli, and k_{ij}^2 the shear correction factors.

The consistent load vector due to the distributed normal load, q, as well as bending

moments, $\tilde{M}_{xx}$, $\tilde{M}_{yy}$, and transverse shear force, $\tilde{Q}$, prescribed at the portion s_τ of the element boundary s may be written as

$$f^{\mathrm{t}} = \int\int_A q\{\boldsymbol{\xi}, \boldsymbol{L}, \boldsymbol{M}\}\,\mathrm{d}A + \int_{s\cap s_\tau} \{0, \tilde{M}_{yy}\boldsymbol{\xi}, \tilde{M}_{xx}\boldsymbol{\xi}\}\,\mathrm{d}s + \int_{s\cap s_\tau} \tilde{Q}\{\boldsymbol{\xi}, \boldsymbol{L}, \boldsymbol{M}\}\,\mathrm{d}s\,. \tag{5.2}$$

The element bending moment and transverse shear stress resultants may be determined from the relations

$$\begin{Bmatrix} M_{xx} \\ M_{yy} \\ M_{xy} \end{Bmatrix} = \boldsymbol{D}_{\mathrm{b}}\boldsymbol{B}_{\mathrm{b}}\boldsymbol{\delta}\,, \tag{5.3}$$

$$\begin{Bmatrix} Q_x \\ Q_y \end{Bmatrix} = \boldsymbol{G}_{\mathrm{s}}\boldsymbol{B}_{\mathrm{s}}\boldsymbol{\delta}\,, \tag{5.4}$$

where the element nodal dof vector has the form

$$\boldsymbol{\delta}^{\mathrm{t}} = \{\boldsymbol{w}^{\mathrm{t}}, \boldsymbol{\theta}_x^{\mathrm{t}}, \boldsymbol{\theta}_y^{\mathrm{t}}\}\,. \tag{5.5}$$

It is noted once again that element matrices (5.1), (5.2) are exactly integrated in an explicit manner by the use of (4.3).

6. Shear correction factor

The notion of the element-appropriate shear correction factors, k_{ij}^2, incorporating both classical (through-the-thickness), k_{ij*}^2, and finite element (in-plane), ϕ^2, corrections to the transverse shear stress resultants and energy, was proposed by Tessler and Hughes in [15] (this concept is closely related to Fried's energy balancing [7] and MacNeal's residual bending flexibility [9] techniques). For the kth element in a finite element mesh, such shear corrections are defined as

$$k_{ij(k)}^2 = k_{ij*}^2\phi_{(k)}^2\,, \tag{6.1}$$

where subscripts i and j are used to denote a general anisotropic material case. In the case of isotropy,

$$k_{(k)}^2 = k_*^2\phi_{(k)}^2\,. \tag{6.1a}$$

This concept is founded on the assertion that the element transverse shear strains can exhibit spurious constraining caused either by assumed kinematic interpolations or by boundary restraints. Therefore, it appears advantageous to complement the classical shear corrections with the appropriate finite element factors, as defined by (6.1).

The classical factors are generally found analytically. For example, Mindlin [19] used a dynamic approach by matching the plate frequency of plane-wave thickness-shear motion for an infinitely long wavelength with that of three-dimensional elasticity. In the present effort, however, we shall only be concerned with the element factors, $\phi^2_{(k)}$, assuming that the k^2_{ij*} are given.

In [15], the authors employed an explicit deflection matching procedure to determine suitable $\phi^2_{(k)}$-factors for a two-node Timoshenko beam element, T1CC4, and a four-node quadrilatieral, MIN4. Herein, we propose an analogous but a somewhat more general approach.

Consider a boundary-value problem for a linearly elastic, shear-deformable (Mindlin theory) plate. Furthermore, for simplicity and clarity of the presentation, assume that no initial strains or stresses exist. Upon discretization of the plate domain with, say, N elements, which employ shear correction factors defined in (6.1), the solution for the total potential energy may be found as

$$\Pi = \sum_{k=1}^{N} \Pi_{(k)} = -\sum_{k=1}^{N} U_{(k)} = -\sum_{k=1}^{N} (U^{\mathrm{b}}_{(k)} + \phi^{-2}_{(k)} U^{\mathrm{s}}_{(k)}), \tag{6.2}$$

where $U^{\mathrm{b}}_{(k)}$ and $\phi^{-2}_{(k)} U^{\mathrm{s}}_{(k)}$ represent the bending and transverse shear strain energies for the kth element, and $\phi^{-2}_{(k)}$ are yet unknown.

Now consider an exact total potential energy solution, Π^*, obtained with an identical mesh but by means of higher-order ('exact') elements that require only the classical shear corrections. This solution will have an analogous form:

$$\Pi^* = \sum_{k=1}^{N} \Pi^*_{(k)} = -\sum_{k=1}^{N} U^*_{(k)} = -\sum_{k=1}^{N} (U^{\mathrm{b}*}_{(k)} + U^{\mathrm{s}*}_{(k)}), \tag{6.3}$$

where superscript * denotes the exact quantities.

Apparently, the approximate solution (6.2) can be improved by way of $\phi^2_{(k)}$-factors, which may enable further minimization of the potential energy. To achieve this, simply set

$$\Delta = \Pi - \Pi^* \equiv 0, \tag{6.4}$$

which results in

$$-\sum_{k=1}^{N} (U^{\mathrm{b}}_{(k)} + \phi^{-2}_{(k)} U^{\mathrm{s}}_{(k)}) + \sum_{k=1}^{N} (U^{\mathrm{b}*}_{(k)} + U^{\mathrm{s}*}_{(k)}) \equiv 0 \tag{6.4a}$$

or, in an alternative form,

$$\sum_{k=1}^{N} (U^{\mathrm{b}*}_{(k)} - U^{\mathrm{b}}_{(k)} + U^{\mathrm{s}*}_{(k)} - \phi^{-2}_{(k)} U^{\mathrm{s}}_{(k)}) \equiv 0 \tag{6.4b}$$

Condition (6.4b) should be enforced for every element, which yields N independent equations in terms of $\phi^{-2}_{(k)}$, i.e.,

$$U^{b*}_{(k)} - U^{b}_{(k)} + U^{s*}_{(k)} - \phi^{-2}_{(k)} U^{s}_{(k)} \equiv 0 \tag{6.5}$$

Solving (6.5) for $\phi^2_{(k)}$ gives

$$\phi^2_{(k)} = \left\{ \left(\frac{U^{s*}}{U^{s}} \right)_{(k)} + \left(\frac{U^{b}}{U^{s}} \right)_{(k)} \left[\left(\frac{U^{b*}}{U^{b}} \right)_{(k)} - 1 \right] \right\}^{-1} \quad (k = 1, 2, \ldots, N). \tag{6.6}$$

Assuming proportionality of the exact and approximate element strain energies, one can write

$$C^{s}_{(k)} = \left(\frac{U^{s*}}{U^{s}} \right)_{(k)}, \qquad C^{b}_{(k)} = \left(\frac{U^{b*}}{U^{b}} \right)_{(k)}, \tag{6.7}$$

where $C^{s}_{(k)}$ and $C^{b}_{(k)}$ are proportionality coefficients, and

$$\psi^2_{(k)} = \left(\frac{U^{b}}{U^{s}} \right)_{(k)} \tag{6.8}$$

is the kth element material-geometric parameter. Then (6.6) takes the form

$$\phi^2_{(k)} = [C^{s}_{(k)} + (C^{b}_{(k)} - 1)\psi^2_{(k)}]^{-1}. \tag{6.9}$$

Equation (6.9), once introduced into (6.2), will guarantee complete potential energy convergence for any discretization. Note that coefficients $C^{s}_{(k)}$ and $C^{b}_{(k)}$ may be regarded as being problem and element dependent. However, $\psi^2_{(k)}$ is a material-geometric parameter which is strictly element dependent. In fact, as will be demonstrated in the beam case, $\psi^2_{(k)}$ is nothing more but the element stiffness penalty parameter.

To achieve a problem invariant character for $\phi^2_{(k)}$, we simply replace $C^{s}_{(k)}$ and $C^{b}_{(k)}$ with permanent element constants C^{s} and C^{b}, respectively. This yields

$$\phi^2_{(k)} = [C^{s} + (C^{b} - 1)\psi^2_{(k)}]^{-1}. \tag{6.10}$$

Introducing (6.10) into (6.4) results in the error

$$\Delta_N = \Pi - \Pi^* = U^{s*}\varepsilon^{s}_{N} + U^{b*}\varepsilon^{b}_{N}, \tag{6.11}$$

with

$$\varepsilon^{b}_{N} = 1 - \frac{\tilde{U}^{b}}{U^{b*}}, \qquad \varepsilon^{s}_{N} = 1 - \frac{\tilde{U}^{s}}{U^{s*}}, \tag{6.11a}$$

where $\tilde{U}^{b}_{N}$ and $\tilde{U}^{s}_{N}$ are the improved bending and shear energy values given by

$$\tilde{U}^{b} = C^{b} \sum_{k=1}^{N} U^{b}_{(k)} = C^{b} U^{b}, \qquad \tilde{U}^{s} = C^{s} \sum_{k=1}^{N} U^{s}_{(k)} = C^{s} U^{s}. \tag{6.12}$$

By establishing C^{s}, $C^{b} \geq 1$, enabling reasonably small error Δ_N for the coarsest mesh, say

$N = 1$, the element behavior is properly tuned. Clearly, the resulting convergence is more rapid than that due to the exclusive application of the classical correction ($C^s = C^b = 1$), i.e.,

$$|\varepsilon^b_N(C^b \geq 1)| \leq |\varepsilon^b_N(C^b = 1)|, \qquad |\varepsilon^s_N(C^s \geq 1)| \leq |\varepsilon^s_N(C^s = 1)| \tag{6.13}$$

and

$$|\Delta_N(C^s, C^b \geq 1)| \leq |\Delta_N(C^s = C^b = 1)|\,. \tag{6.13a}$$

EXAMPLE 6.1 (Beam example). It is instructive to address first the T1CC4 Timoshenko element [24] (its stiffness matrix is identical to the one-point Gauss quadrature linear beam of [8]) and seek an explicit form of (6.10) in a one-dimensional case.

The T1CC4 two-node beam element is a one-dimensional analogue of MIN3. Its kinematic field variables w and θ possess complete quadratic and linear polynomial interpolations, respectively; the mid-span w-dof is constrained out by means of a continuous differential shear relation (à la (4.6)), enforcing a constant shear along the element. The T1CC4 stiffness matrix, corresponding to the nodal dof vector $\{w_1, w_2, \theta_1, \theta_2\}$, may be written as

$$\boldsymbol{K} = \boldsymbol{K}_b + \boldsymbol{K}_s = \alpha_f \left\{ \begin{bmatrix} 0 & 0 & 0 & 0 \\ & 0 & 0 & 0 \\ & & 1 & -1 \\ \text{Sym.} & & & 1 \end{bmatrix} + \alpha \begin{bmatrix} a^2 & -a^2 & a & a \\ & a^2 & -a & -a \\ & & 1 & 1 \\ \text{Sym.} & & & 1 \end{bmatrix} \right\} \tag{6.14}$$

with

$$a = 2/l, \qquad \alpha_f = EI/l$$

and the penalty parameter, α, is given by

$$\alpha = \tfrac{1}{4} k^2 \frac{G}{E} \left(\frac{l}{r}\right)^2 \tag{6.15}$$

(E and G are Young's and shear moduli, l is the element length, I and r are the cross-sectional moment of inertia and radius of gyration, respectively).

Utilizing the appropriate solutions to a cantilever beam/tip load problem (e.g., see solutions in [15]) in (6.7), (6.8) gives

$$C^s = 1, \qquad C^{-1} = C^b - 1 = \tfrac{1}{3}, \qquad \psi^2 = \alpha_* = \tfrac{1}{4} k_*^2 \frac{G}{E} \left(\frac{l}{r}\right)^2 \tag{6.16}$$

and

$$\phi^2 = \frac{1}{1 + \alpha_*/C}\,. \tag{6.17}$$

Substituting (6.17) into (6.15), where $k^2 = k_*^2 \phi^2$ is employed, yields

$$\alpha = \frac{\alpha_*}{1 + \alpha_*/C}\,. \tag{6.18}$$

Clearly, the original T1CC4 stiffness matrix (i.e., for $k^2 = k_*^2$) is inherently ill-conditioned in thin limit ($l/r \to \infty$) owing to the difference in the order of the shear and bending stiffness coefficients, i.e.,

$$\frac{k_{sii}}{k_{bii}} = \alpha_* \to \infty \quad (i = 3, 4). \tag{6.19}$$

On the other hand, ϕ^2 of (6.17) assures a bounded form for α of (6.18) and, consequently, a well-conditioned stiffness over the entire range of the slenderness ratio (l/r). Alternatively, (6.18) may be written in terms of an inverse of α, i.e.,

$$\alpha^{-1} = \alpha_*^{-1} + C^{-1}. \tag{6.20}$$

This linear relation is particularly insightful owing to the direct proportionality of α^{-1} and the element flexibility. The improved, α^{-1}, and the original, α_*^{-1}, flexibilities simply differ by a constant shift factor C^{-1}. By increasing C^{-1}, the solution becomes more flexible. Conversely, by lowering C^{-1}, we decrease the flexibility, α^{-1}, and achieve stiffer solutions. In the limiting case, $C^{-1} = 0$, the original form is achieved.

MIN3 shear correction.

To obtain a suitable ϕ^2-factor for MIN3, one can immediately generalize (6.17) in the form

$$\phi^2 = \frac{1}{1 + \alpha(k_{ij*}^2)/C}, \tag{6.21}$$

where C is a permanent constant to be determined by a numerical experiment (Section 8), and

$$\alpha(k_{ij*}^2) \overset{\text{df}}{=} \frac{\sum_{i=4}^{9} k_{sii}}{\sum_{i=4}^{9} k_{bii}} \propto \frac{A}{h^2} \tag{6.22}$$

is a generalized penalty parameter defined as a ratio of the diagonal shear and bending stiffness coefficients associated with the rotational degrees of freedom. (If we were to degenerate MIN3 into a two-node Timoshenko beam element, the definition of (6.22) would come out the same as that of (6.16) for the beam element.)

This form of α, tested numerically along with several other similar definitions, yielded an overall superior element behavior. Therefore, (6.22) will constitute the basis for MIN3's shear correction factor.

7. Boundary condition related kinematic overrestraining

In this section, a boundary condition related shear locking of a two-node Timoshenko element (T1CC4 [24]) that satisfies Criterion 3.1 is demonstrated.

Let us examine a one-element solution for a beam loaded by a moment M_2 at node 2 and subjected to the boundary conditions

$$w_1 = w_2 = \theta_1 = 0\,. \tag{7.1}$$

The solution to the only unknown degree-of-freedom, θ_2, is simply

$$\theta_2 = 4\theta_{2(\text{exact})}\left[1+\frac{k^2G}{4E}\left(\frac{l}{r}\right)^2\right]^{-1}, \tag{7.2}$$

where $\theta_{2(\text{exact})}$ denotes an exact Bernoulli–Euler theory solution given by

$$\theta_{2(\text{exact})} = \frac{M_2 l}{4EI}\,. \tag{7.2a}$$

The bending curvature, K, and shear strain, γ, are constant along the element (owing to the interpolation assumptions), and proportional to θ_2 for this problem, i.e.,

$$K = -\frac{2}{l}\theta_2, \qquad \gamma = -\tfrac{1}{2}\theta_2\,. \tag{7.3}$$

Evidently, with the classical shear correction factor, $k^2 = k_*^2 = \frac{1}{12}\pi^2$, solutions (7.2), (7.3) become worthless in the thin regime, i.e.,

$$\theta_2, K, \gamma \to 0 \qquad \text{as } \frac{r}{l}\to 0 \quad \text{(locking)}\,.$$

It is noted, however, that T1CC4 will not lock under regular boundary restraints, i.e., when any two restraints are imposed on w and θ to delete the rigid body motion.

Interestingly, the element-appropriate shear correction factor approach (Section 6) can overcome this pitfall. Indeed, replacing the shear correction factor in (7.2) with that of (6.1a), (6.16), (6.17) yields the exact solution for θ_2, i.e.,

$$\theta_2 = \frac{4\theta_{2(\text{exact})}}{1+3\left[1+\dfrac{12E}{k_*^2 G}\left(\dfrac{r}{l}\right)^2\right]^{-1}} \xrightarrow{r/l\to 0} \theta_{2(\text{exact})}\,. \tag{7.4}$$

Moreover, exact values of the shear force, bending moment at the clamped end, and the total potential energy are recovered.

The problem of the boundary condition related locking, obviously trivial in the beam situation and completely avoidable by using more than one element, may become significant when plates are restrained. In Section 8, we will comment further on a possible boundary overrestraining of MIN3 elements when used with exclusively classical shear correction factors.

8. Numerical studies

Several numerical experiments dealing with isotropic, homogeneous plates are presented. At the outset, employing the classical shear correction factor alone (i.e., $k^2 = \frac{1}{12}\pi^2$), MIN3's kinematic reliability is examined (i.e., stiffness matrix rank). Then follow patch tests for bending and transverse shear, and a convergence study of thin square plates.

The subsequent analyses employ the element-appropriate shear correction factor of (6.1a), (6.21), (6.22) that replaces the classical correction in the transverse shear expressions. First, an optimal value of constant C is established, and using this shear correction, the element performance is assessed on square, rectangular, circular and rhombic plates under various loadings and boundary conditions.

In all studies involving uniform loading, MIN3's consistent load vector is used. The bending moments and transverse shear force resultants are evaluated at the element centroid (note that bending moments are constant throughout the element).

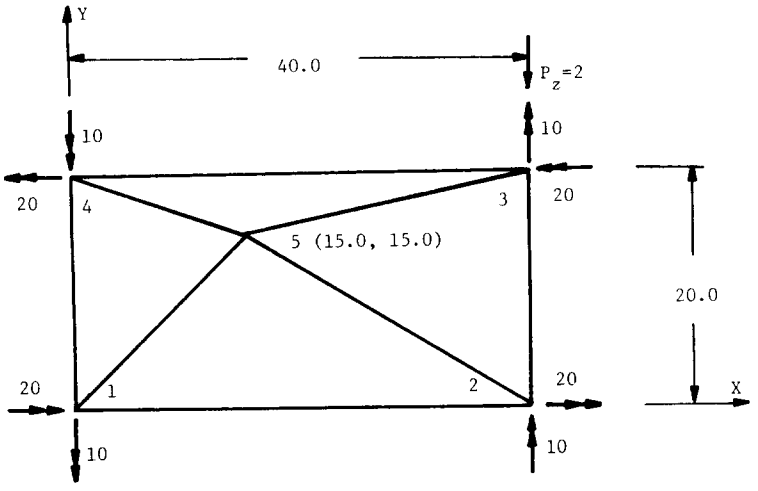
As a consequence of a skew distortion of quadrilateral elements, even generally well-performed elements exhibit a stiffening effect (see [16]). For this reason, our latest quadrilateral, MIN4 [15], is subjected to the rhombic plate tests reported in [16].

Spectral analysis of stiffness matrix. Upon performing a spectral analysis of MIN3's stiffness matrix, three zero eigenvalues associated with the requisite rigid body modes are revealed. No spurious zero energy modes are present, as one would expect from an exactly integrated stiffness matrix. Hence, the stiffness matrix is of a correct rank, and the element should be regarded as kinematically reliable.

Kirchhoff patch test. The popular 'Kirchhoff plate' patch test [27] is undertaken (refer to Fig. 4). A thin, rectangular plate is subjected to the state of bending such that all three moment resultants are constant throughout the plate domain. The plate discretization consisting of four geometrically distinct MIN3 elements yielded exact values of all three constant moments in each of the elements.

Shear patch test. Since the present refined theory involves transverse shear stress resultants, it may be desirable to conduct an appropriate shear patch test. To do this, we propose to examine an infinitely long clamped plate subjected to uniform transverse line load, Q_z, along the long free edge (see Fig. 5(a)). Here, the state of constant Q_{xz}-shear exists (i.e., $Q_{xz} = Q_z$), while Q_{yz} is zero everywhere in the plate. The narrow strip four-element model subject to symmetric boundary conditions (Fig. 5(b)) produced exact values of Q_{xz} and Q_{yz} in each of the elements.

Thin square plates. An analysis is carried out for a simply supported, thin square plate under uniform load (plate span-to-thickness ratio $L/h = 100$). Solutions are obtained for three distinct mesh patterns (refer to Fig. 6) and compared with those of the exact quadrature triangle of Hughes and Taylor [12]. The center deflection and bending moment results are summarized in Table 1. Evidently, MIN3's performance is superior, and particularly, for A and B discretizations for which the Hughes–Taylor solutions are very stiff. However, for thinner plates ($L/h > 100$) and A and B meshes, MIN3 produces considerably stiffer solutions (e.g., MIN3 values for $L/h = 1000$ closely resemble those of Hughes and Taylor for $L/h = 100$). This stiffening effect may perhaps be attributed to an overrestraining of the kinematic field by



DATA: E = 1000.0, υ = 0.3, h = 1.0

BOUNDARY RESTRAINTS: w = 0 at nodes 1, 2 and 4.

ANALYTIC SOLUTION: $M_{xx} = M_{yy} = M_{xy} = 1.000$ throughout the plate.

Fig. 4 Thin plate patch test.

boundary conditions as discussed in relation to beams in Section 7. These results also suggest that further improvements must be sought. Subsequently, we shall invoke the element-appropriate shear correction factor approach to enhance the element behavior.

Optimal C value. An effective utilization of the element-appropriate shear correction factors requires an optimal choice of C in (6.21). To obtain the appropriate value, we propose a least-squares minimization of the strain energy error associated with the A and B discretizations of a centrally loaded thin square plate (refer to Fig. 6). Both simply supported and clamped boundary conditions are considered. Note that the two mesh patterns produce the same number of elements and degrees-of-freedom per discretization N. It may therefore be meaningful to seek the value of C that yields comparable energy error (in the absolute value sense) for each of the meshes. In Fig. 7, strain energy error curves based on solutions with $C = \frac{7}{4}$, 2 and $\frac{9}{4}$ are plotted versus C. In each boundary condition case, a least-squares fit provides a value of C corresponding to zero energy error. These are $C_{SS} = 1.94$ and $C_{CL} = 2.04$ for simply supported and clamped restraints, respectively, and they may be regarded as optimal values for these problems. A compromise value of $C = 2.0$, which is nearly in the mid-range between C_{SS} and C_{CL}, will be assumed optimal and employed throughout the subsequent analyses.

Thin and moderately thick square plates (k^2 is given by (6.1a), (6.21), (6.22), $C = 2.0$). Convergence studies are carried out for thin and moderately thick square plates with L/h of 1000 and 10, respectively. The boundary conditions, discretizations and loadings are shown in Fig. 6. Convergence plots of the center deflection and center bending moment are presented in Figs. 8 and 9. For the purpose of comparison, the results of DKT (discrete Kirchhoff triangle

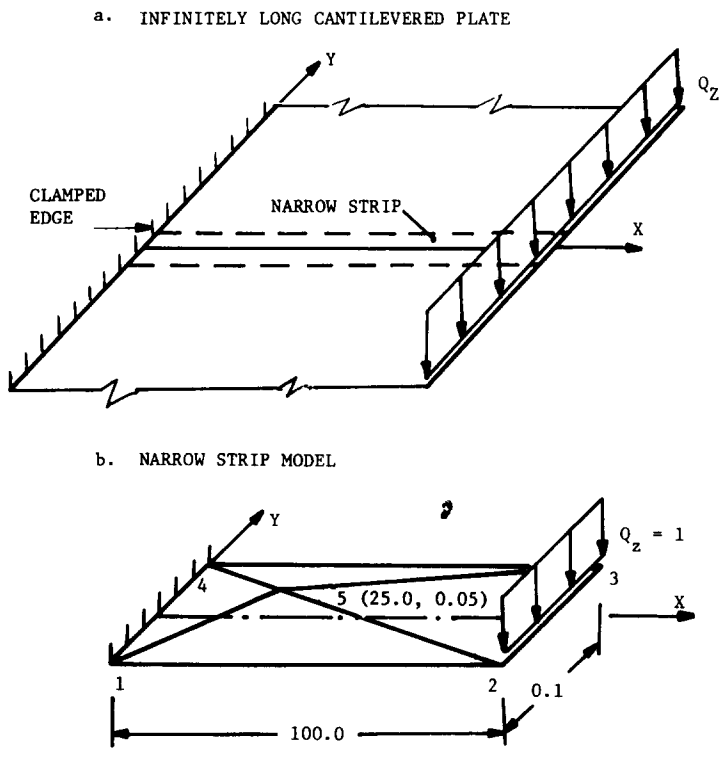
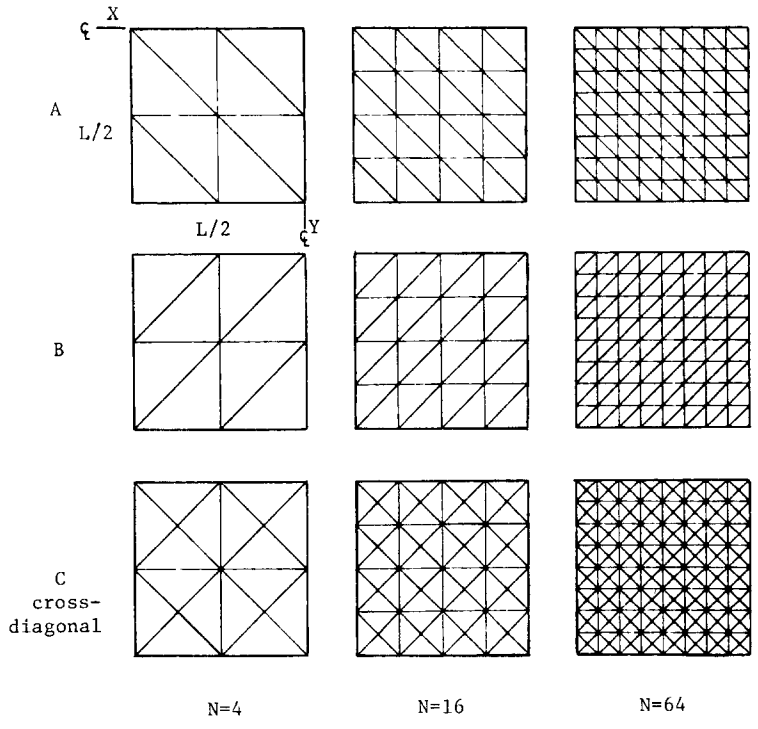


Fig. 5 Transverse shear patch test.

Table 1
Normalized center displacement and bending moment for simply supported, thin square plate subjected to uniform load ($L/h = 100$, $\nu = 0.3$)

	Exact quadrature Hughes–Taylor triangle [12]				MIN3 ($C = \infty$)	
	N	A	B	C	A	B	C
(A) Displacement	4	0.681	0.002	0.912	0.806	0.140	0.957
	16	0.773	0.036	0.978	0.891	0.728	0.989
	64	0.827	0.363	0.994	0.982	0.972	0.997
(B) Bending moment	4	0.555	0.002	0.919	0.651	0.167	0.954
	16	0.539	0.036	0.979	0.766	0.969	0.989
	64	0.580	0.386	0.996	0.953	1.001	0.997



BOUNDARY RESTRAINTS:

SS -- simply supported boundary:

$$w(L/2, y) = w(x, L/2) = \theta_x(L/2, y) = \theta_y(x, L/2) = 0.$$

CL -- clamped boundary:

$$w(L/2, y) = \theta_x(L/2, y) = \theta_y(L/2, y) = 0,$$

$$w(x, L/2) = \theta_x(x, L/2) = \theta_y(x, L/2) = 0.$$

LOADING:

U -- uniform load; C -- center load.

Fig. 6. Square plate; symmetric quadrant mesh, boundary and load conditions.

[10]) are shown in Fig. 8, while MIN4 solutions for the moderately thick plate are depicted in Fig. 9. Clearly, MIN3's performance is excellent in the two plate regimes. The element competes well with the accurate DKT in the simply supported plate case, and for a clamped plate, MIN3's solutions are consistently more accurate than those of DKT.

Thin circular plate. Circular plate bending problems are convenient in assessing an element behavior under arbitrary distorted shapes. The typical discretizations of a symmetric quadrant of a thin circular plate ($2R/h = 100$) are shown in Fig. 10. The nodes on the circular plate boundary are fixed in the transverse displacement direction, i.e., this is a simply supported restraint of type SS1 (see discussion in [28] on alternate simply supported boundary conditions

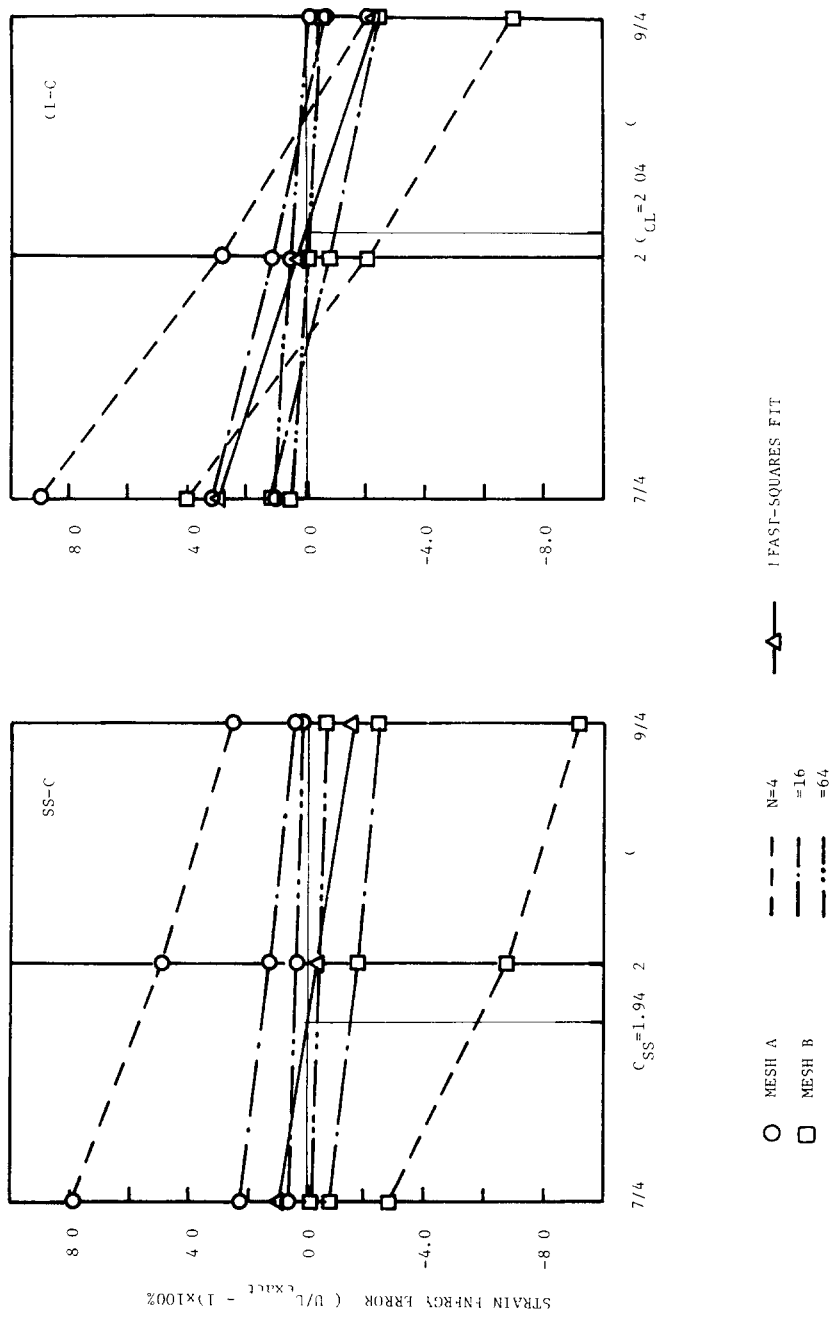


Fig. 7. Simply supported and clamped thin square plates under center load, least-squares approach for optimal C values.

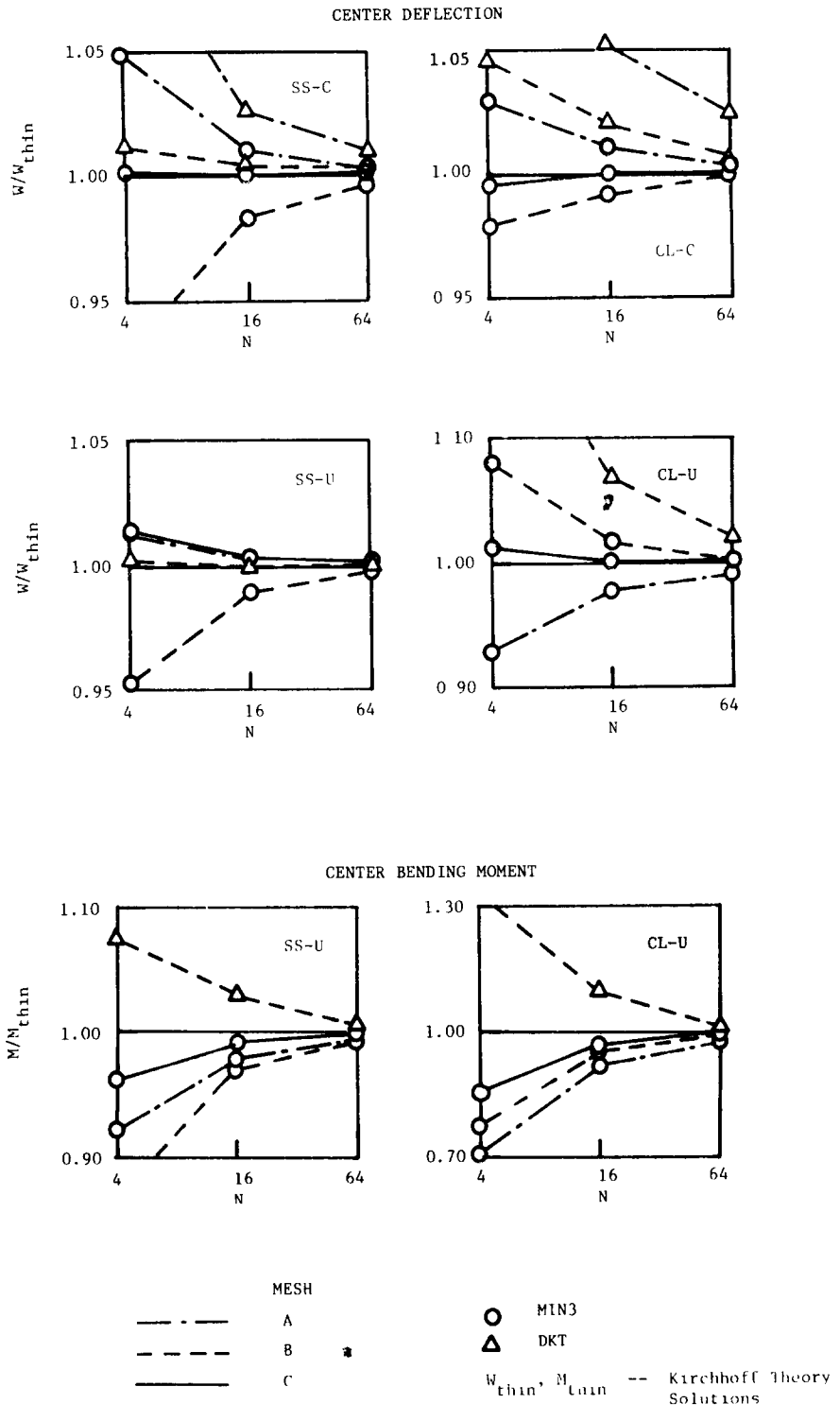


Fig. 8. Convergence study of thin square plates ($L/h = 1000$, $\nu = 0.3$).

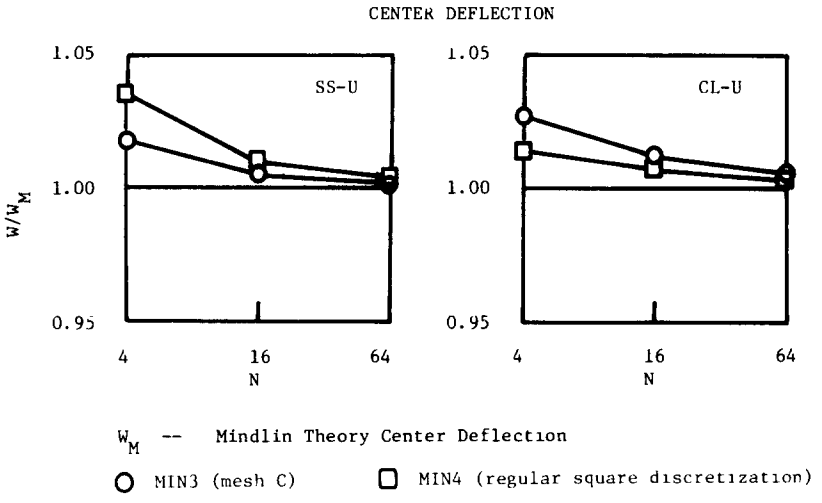


Fig 9 Convergence study of moderately thick square plates ($L/h = 10$, $\nu = 0.3$)

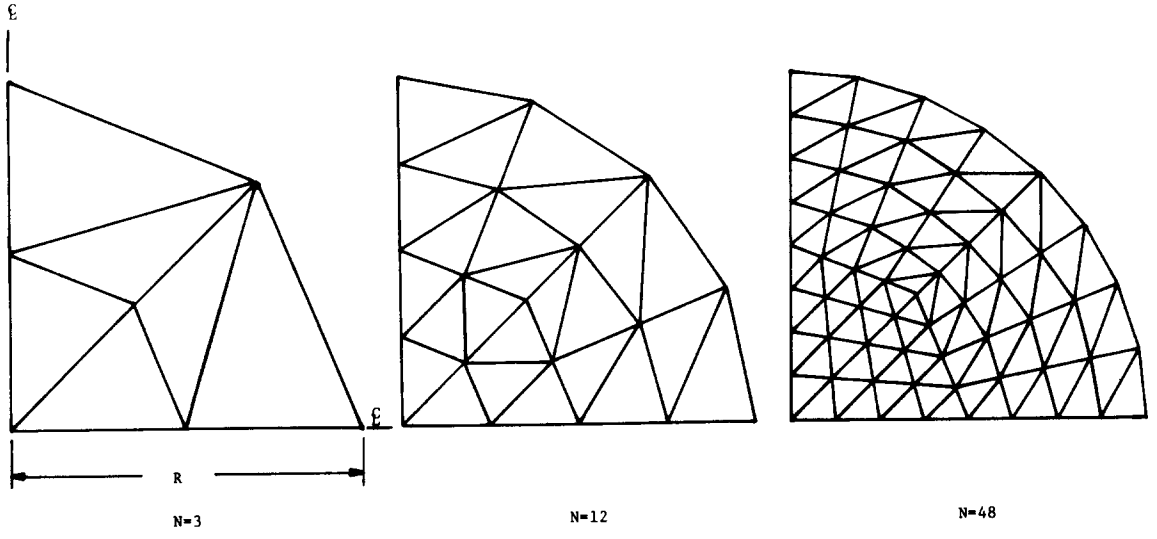


Fig. 10. Symmetric quadrant discretizations of circular plate.

for Mindlin plates). Table 2 contains center deflection and center bending moments results obtained with a one-point centroidal quadrature triangle of Hughes and Taylor [12] and the two MIN3 versions (i.e., the classical shear correction ($C = \infty$) and the element-appropriate shear correction ($C = 2$)). Evidently, MIN3 ($C = 2$) produced consistently superior solutions.

Twisted ribbon tests. A cantilevered, thin rectangular plate subject to a twisting moment at the free end is regarded as a severe test for plate bending elements under large aspect ratio distortions [29]. Herein, two mesh patterns are tried: a cross-diagonal four-element mesh (A) and a two-element mesh (B) (refer to Figs. 11 and 12). The tip deflection results are compared

Table 2
Normalized center displacement and bending moment for simply supported, thin circular plate subjected to uniform load ($2R/h = 100$, $\nu = 0.3$)

	N	1-point centroidal quadrature Hughes–Taylor triangle [12]	MIN3	
			$C = \infty$	$C = 2$
(A) Displacement	3	0.703	0.859	1 058
	12	0.912	0.953	1.014
	48	0.948	0.990	1 003
(B) Bending moment	3	0.576	0.720	0.910
	12	0.878	0.915	0.980
	48	0.975	0.984	0 994

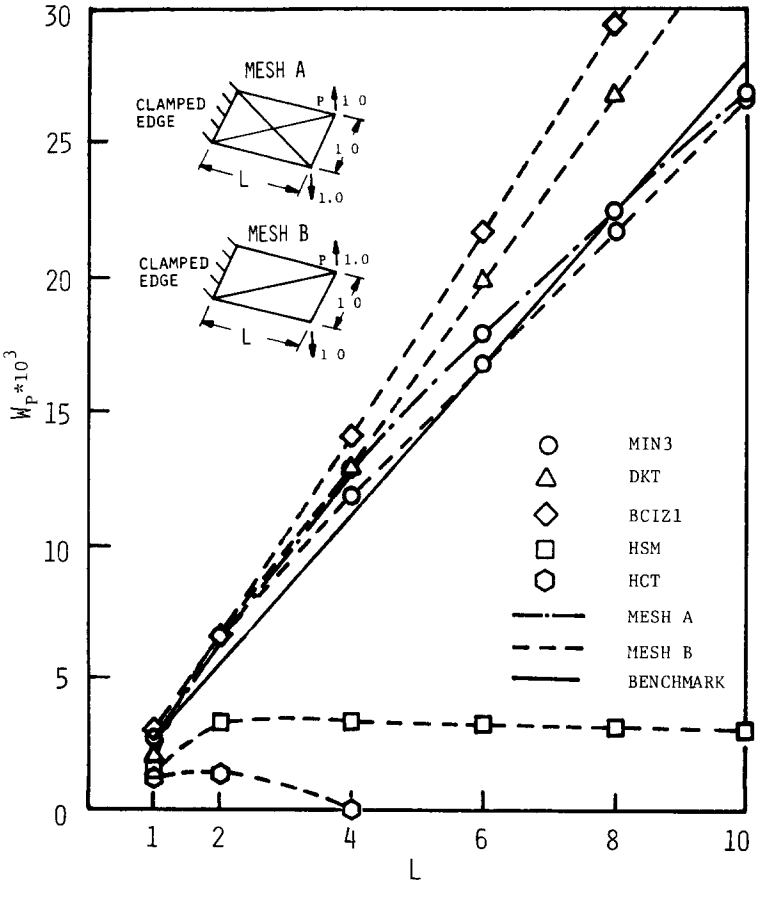


Fig. 11. Twisted ribbon test via transverse corner forces ($E = 10^7$, $\nu = 0.25$, $h = 0.05$).

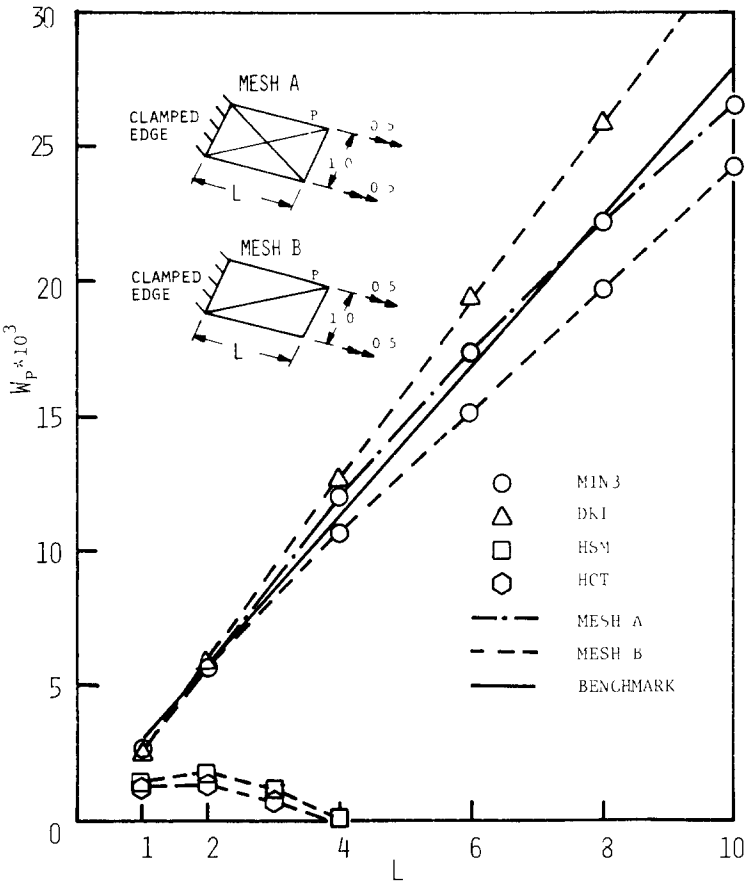


Fig. 12. Twisted ribbon test via twisting corner moments ($E = 10^7$, $\nu = 0.25$, $h = 0.05$).

with the benchmark solutions [29], and the results produced by the popular nine degrees-of-freedom, thin triangles, namely, BCIZ1 (a nonconforming displacement model of Bazeley et al. [2]), HSM (a hybrid stress model of Allwood and Cornes [4]), HCT (a conforming displacement model of Clough and Tocher [1]) and DKT. These results were taken from [13]. In this comparison again, MIN3 appears to outperform the other triangular elements.

Thin rhombic plates. A thin, simply supported rhombic plate under uniform load (refer to Fig. 13) is a challenging test problem for plate bending elements. Here, according to Morley's analytic solution [30], bending moments M_{xx} and M_{yy} at the obtuse plate corner possess a singular character, and moreover, they have opposite signs. Many thin elements exhibit pathological results that are manifested by moments of the same sign, and several reduced integration quadrilaterals produce somewhat oscillatory solutions for this problem (see [31, 32]). In Fig. 13, along with the plate configuration, two quarter-plate meshes comprised of ninety-six quadrilateral elements, MIN4 and T1 (a four-node, Mindlin element based on an assumed shear strain interpolation [11]), and sixty-four triangular MIN3 elements are illustrated. The moment results, corresponding to the SS1 boundary restraints (i.e., $w = 0$ along

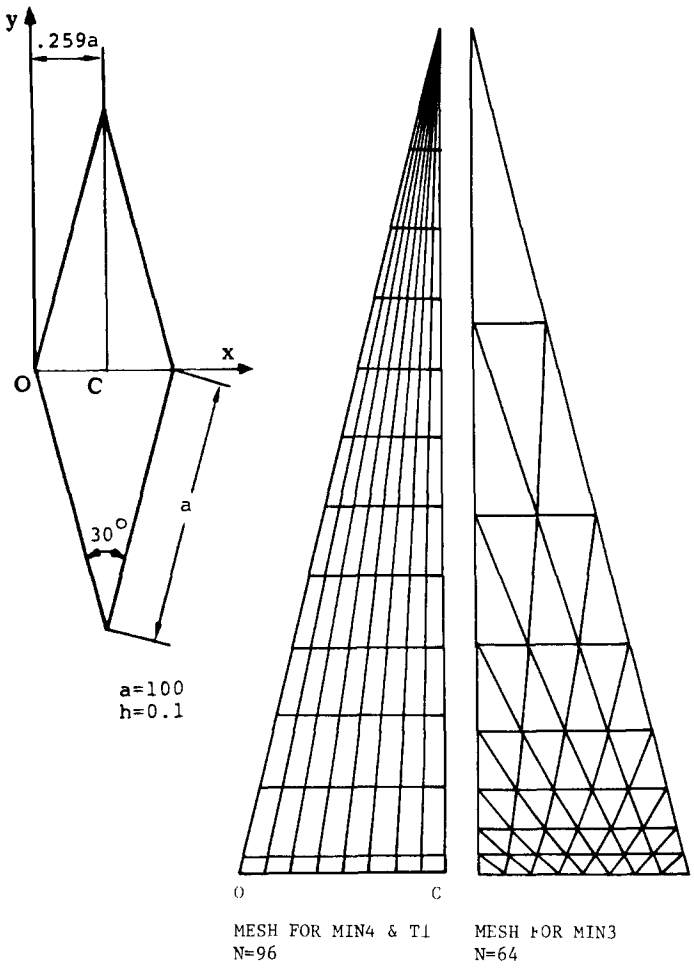


Fig. 13. Simply supported, thin rhombic plate under uniform load, quadrilateral and triangular element discretizations of symmetric quadrant.

the boundary) are depicted in Fig. 14. Note that the analytic solution trend is captured well by all three elements; however, it appears that MIN3's results are somewhat superior.

Crisfield [16] demonstrated how even generally well-behaved quadrilaterals exhibit excessive stiffness under skew distortions. His results for uniformly loaded thin rhombic plates (see the discretizations and boundary restraints in Fig. 15 and Table 3) are shown in Figs. 16 and 17, where the following elements were tested: T1, T (a four-node, thin element based on shear constraints [16]) and S8 (an eight-node, uniform/reduced integration Mindlin element [33]). Also depicted in these plots are solutions obtained with MIN4. It is clear from the comparison that MIN4's performance under skew distortions is superior to the other elements.

Remarks on transverse shear resultants

Many three- and four-node shear-deformable plate and shell elements are known to yield inferior transverse shear force solutions. Because of this pitfall, these results are rarely reported in the literature and are often disregarded in the applications.

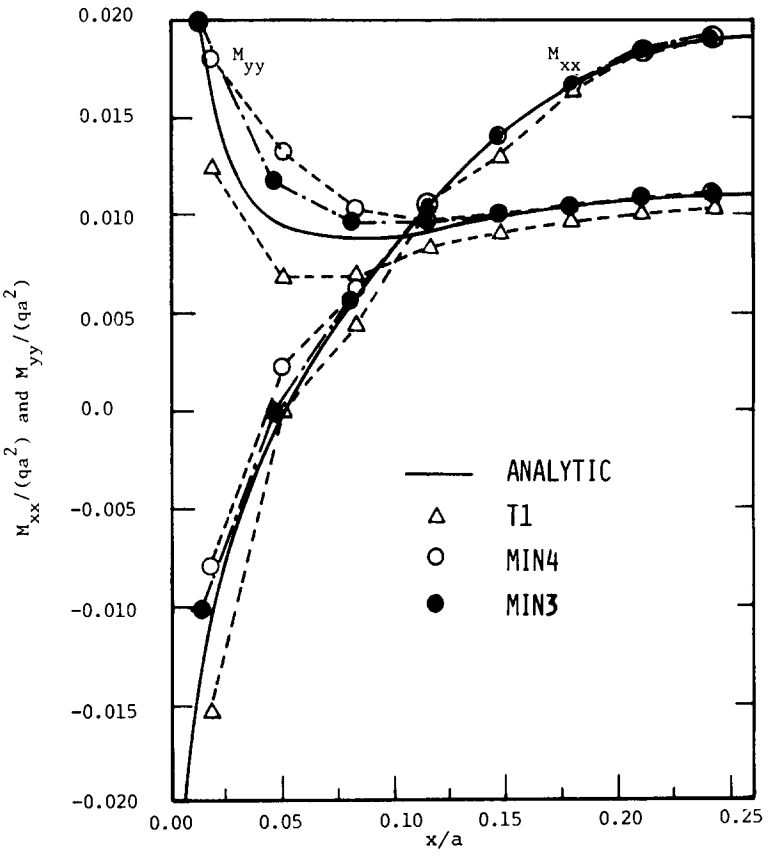
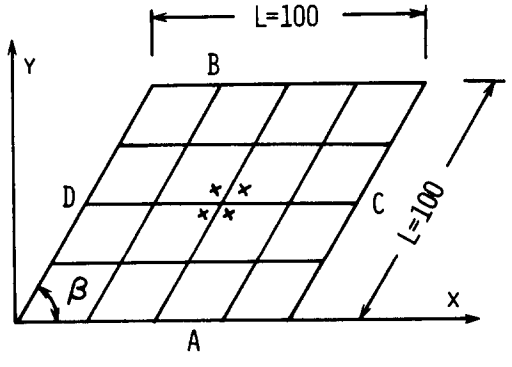


Fig. 14. Simply supported, thin rhombic plate under uniform load, normal bending moments along center line OC.



× 2x2 GAUSS QUADRATURE POINTS NEAREST PLATE CENTER

β = 30° & 60°, THICKNESS = 0.1, N = 16.

Fig. 15. Rhombic plate discretization by skewed quadrilateral elements

Table 3
Boundary conditions for thin rhombic plates

Angle β	Edge boundary restraints A B	C D	Boundary condition designation
60°	$w = 0$	free	BC1
	$w = \theta_y = 0$	free	BC2
30°	$w = 0$		BC3
	$w = \theta_y = 0$	$w = 0$	BC4

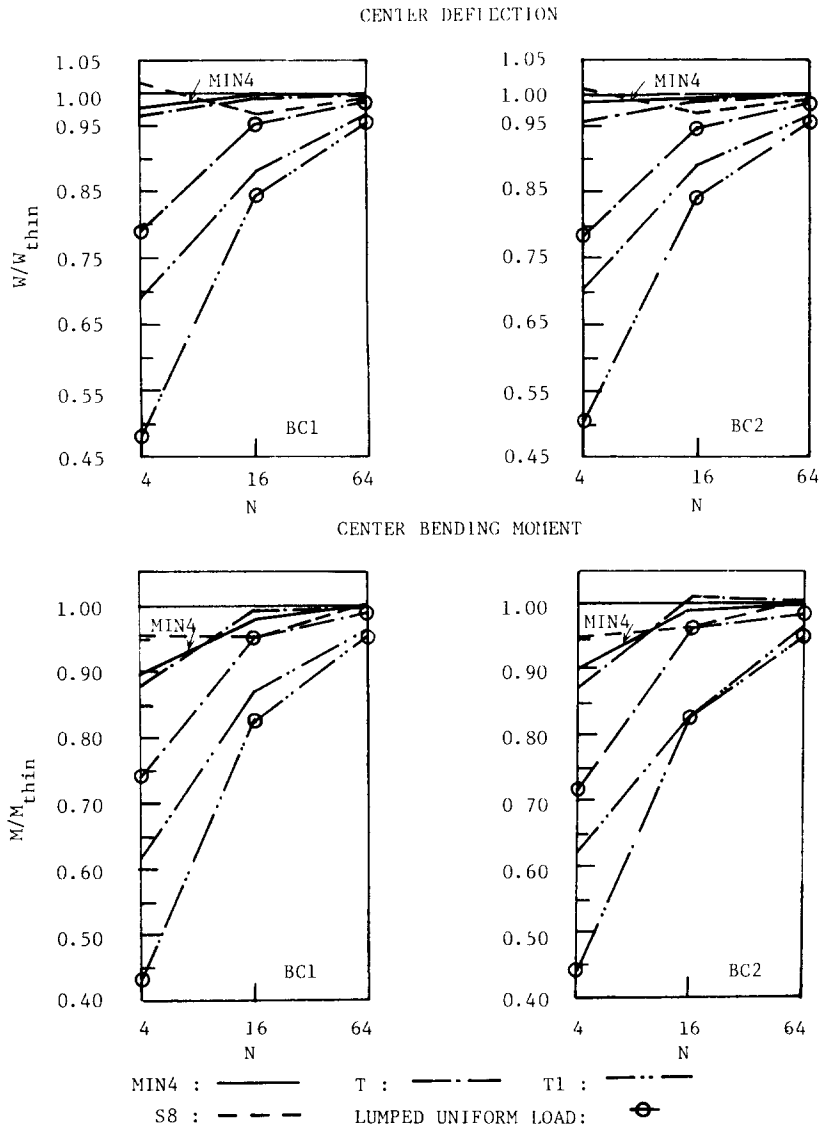


Fig. 16. Convergence study of thin rhombic plate under uniform load discretized with skewed quadrilateral elements ($\beta = 60°$, $\nu = 0.31$).

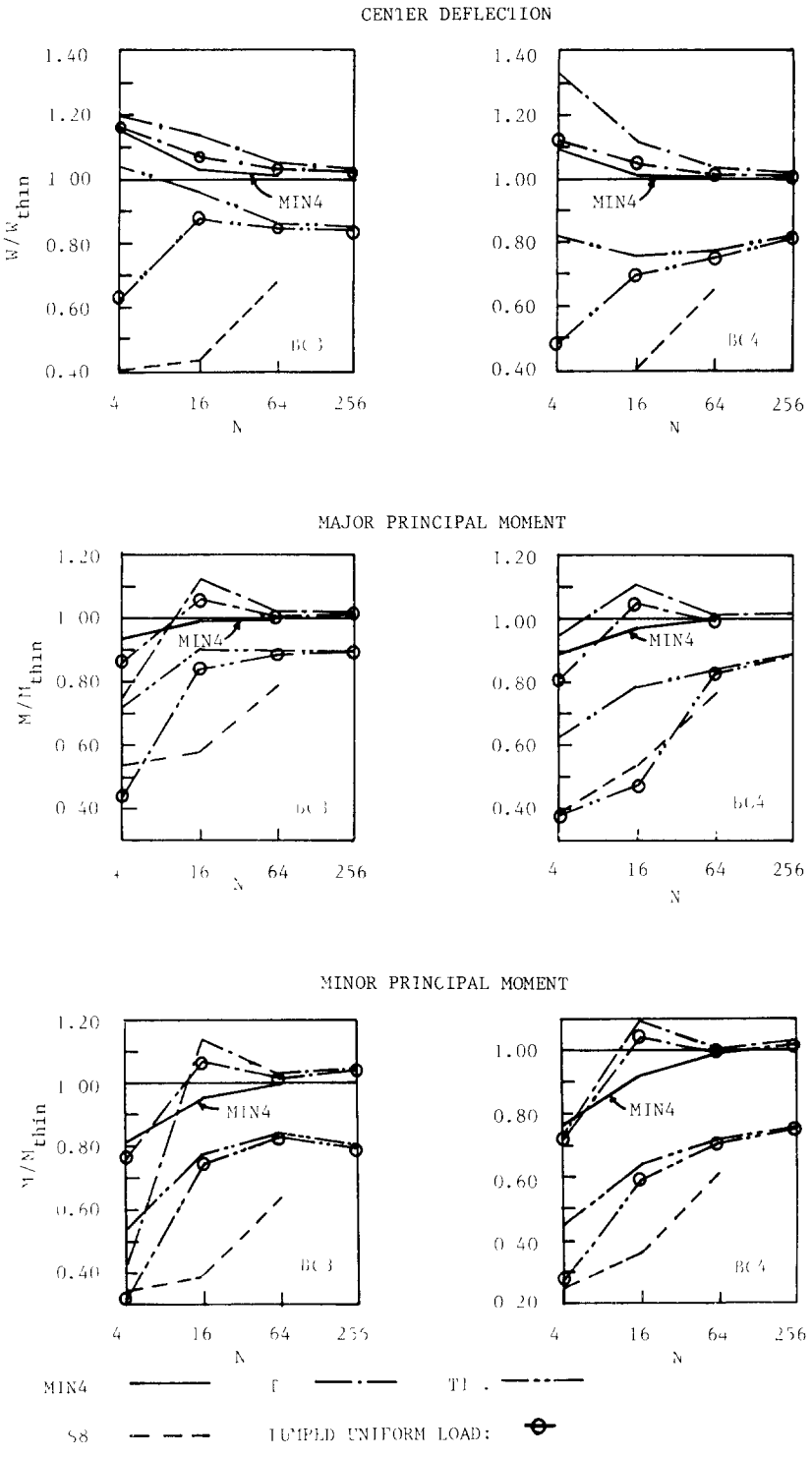


Fig. 17. Convergence study of thin rhombic plate under uniform load discretized with skewed quadrilateral elements ($\beta = 30°$, $\nu = 0.3$)

Our recent four-node quadrilateral plate element (MIN4) evolved as an exception to this trend in that it produces reasonably accurate solutions for the transverse shear resultants. Hence, MIN3 is expected to perform analogously to its four-noded counterpart. In Fig. 18 we present convergence curves of the maximum shear resultants for simply supported square and circular plates.

9. Concluding remarks

The so-called *anisoparametric* methodology, founded on two auxiliary kinematic field criteria, has demonstrated to be effective for the development of a simple, three-node Mindlin element designated herein as MIN3. The approach relies on an exact explicit integration of all element matrices and, hence, assures a consistent and kinematically reliable (correct rank) model. It takes advantage of an explicit degree-of-freedom reduction technique, achieved via 'continuous' shear edge constraints, which generates a compatible kinematic field and a simple, isoparametric-like nodal pattern. There exists no ambiguity regarding derivation of consistent load vectors and, in dynamics, consistent mass matrices (the mass matrices have not been presented herein).

Much success of this formulation is due to the innovative element-appropriate shear correction factor technique, which is responsible for a well-conditioned element stiffness matrix over the entire range of thickness-to-length ratios. Consequently, the element suffers no deficiencies in either the classical (thin) or shear-weak (thick) flexure regimes and at the same time produces rapidly convergent solutions.

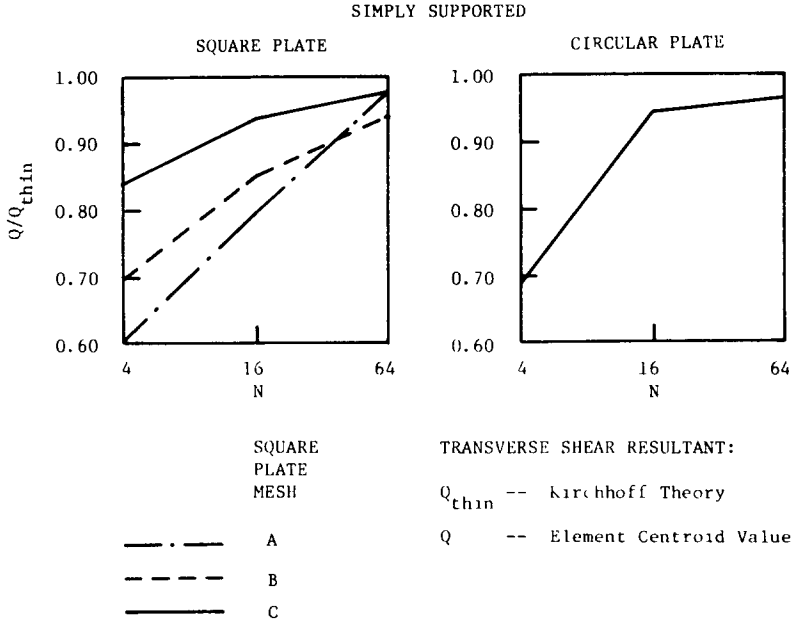


Fig. 18 Convergence of maximum shear resultants for simply supported square and circular plates ($L = 2R = 10$, $h = 0.01$, $\nu = 0.3$)

Based upon extensive numerical testing in the linear, elasto-static regime, we conclude that MIN3 is an excellent element. Due to its reliability, economy, and good stress recovery, it may be regarded as a viable candidate for extention to shell, laminated composite and nonlinear analyses.

Finally, it is important to point out that the recently developed anisoparametric, lower order elements, namely, the two-node Timoshenko beam, T1CC4 [15, 24], a four-node quadrilateral plate, MIN4 [15], and the present three-node triangle plate, MIN3, constitute a family of fully compatible elements that can effectively model girder-reinforced plates.

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